SMART EV TO EV CHARGING WITH LI-FI BASED TECHNOLOGY

INDEX

ABSTRACT

The current boost in electric vehicle (EV) adoption requires better charging systems that can adapt and function more effectively than standard charging systems. Actual situations which include highways and remote locations and emergency response areas have restricted access to permanent charging stations. The project introduces its solution through the development of a vehicle-to-vehicle (V2V) wireless power transfer system which enables one electric vehicle (EV) to transfer power to another EV without requiring direct contact between them.

The power transmitting vehicle and receiving vehicles are designed and charge transfer between these vehicles is done through Inductive coupling using Li-Fi communication. The proposed system uses resonant inductive coupling as its primary mechanism which enables electrical energy to move between two coils that operate at their shared resonant frequency through a high-frequency magnetic field. A MOSFET- based push-pull self-oscillating inverter creates a high- frequency signal which powers the primary coil and generates a powerful magnetic field for the transmitting unit. The receiving coil which operates within effective coupling distance receives the magnetic field and transforms it into electrical power which undergoes rectification to charge batteries.

The system needs careful design choices about coil geometry and number of turns and wire gauge and resonance tuning to achieve efficient power transfer. The system needs resonance compensation to maintain operational efficiency because it functions with loosely connected components which create small air gaps. The design enables two electric vehicles which move independently to transfer energy without making physical contact with each other. The system enables power transfer through the use of Li-Fi technology which uses light as its transmission medium.

LIST OF FIGURES

ACRONYMS

S. No.	Abbreviation	Full Form
1	DC	Direct Current
2	IR	Infrared Radiation
3	VCO	Voltage Controlled Oscillator
4	MOSFET	Metal Oxide Semiconductor Field-Effect Transistor
5	SWG	Standard Wire Gauge
6	EV	Electric Vehicle
7	Li-Fi	Light Fidelity
8	AC	Alternating Current
9	IC	Integrated Circuit
10	PCB	Printed Circuit Board
11	LED	Light Emitting Diode
12	LCD	Liquid Crystal Display

CHAPTER 1
INTRODUCTION

1.1 Background

Electric vehicles have been a subject of research and development for several decades. The earliest concepts of EVs date back to the late 19th century, but it was only in the early 2000s that modern lithium-ion battery technology made them practically viable for everyday use. Since then, companies and research institutions have invested heavily in improving battery capacity, reducing charging time, and expanding charging infrastructure.

Researchers have explored various forms of mobile charging, including robotic charging arms, drone-based delivery of portable batteries, and inductive wireless power transfer between parked vehicles. However, most of these approaches are either too expensive, mechanically complex, or dependent on precise alignment and external coordination systems. The need for a simpler, smarter, and more communication-driven approach remained open.

Li-Fi technology, first demonstrated by Professor Harald Haas in 2011, opened a new direction in wireless communication. Its ability to transmit data through light without any radio frequency dependency made it attractive for environments sensitive to electromagnetic interference. Over the years, researchers explored Li-Fi in hospitals, aircraft cabins, underwater systems and many but its application in vehicle-to-vehicle communication remained largely unexplored.

Motivated by these gaps, we took up this project to combine two evolving fields, EV energy management and Li-Fi-based communication into a single working system. We studied existing literature on peer-to-peer charging, optical wireless communication, and embedded control systems, and designed a prototype where a donor EV detects a low-battery request from a nearby receiver EV through Li-Fi and initiates smart charging automatically. Our work builds on the foundation laid by previous researchers while introducing a practical, low-cost, and interference-free communication layer that makes vehicle-assisted charging a realistic possibility.

1.2 Motivation and Problem Statement

One of the real problems with electric vehicles today is what happens when the battery runs out on the road. With a petrol vehicle, someone can bring a can of fuel and the driver is back on the road in minutes. With an electric vehicle, there is no such option. The driver has to wait for a tow truck or sit idle until a mobile charging unit arrives, and even then, getting a useful amount of charge into the battery takes time. This is critical in areas with no charging stations or in areas where charging stations are few and far between.

The idea behind this project is simple: if one vehicle with a strong battery is already travelling near a vehicle with a weak battery, one can use its battery. No need for any cables, no need for either vehicle to stop, and no need for any special infrastructure on the road. The rear vehicle just follows the front one closely and the energy moves across wirelessly while both vehicles are running. The front vehicle gets enough charge to reach the next charging station, and the whole rescue operation happens without any delay or drama.

Electric vehicle that receives the power and an electric vehicle that transmits this power or charge are built. Then, both the systems are integrated for testing and analysis. Li-Fi is the main source of communication between the electric vehicles which then allow vehicles to charge wirelessly [1].

1.3 Overview of the System

The complete system consists of two small electric vehicles that work together to transfer energy wirelessly while both are in motion. One vehicle carries a weak battery and needs charge. The other carries a strong battery and sends charge across to the first. Neither vehicle stops. No cable connects them. The entire operation happens automatically once the front vehicle detects that its battery is running low.

The front vehicle runs on an 89C52 microcontroller that continuously monitors its 8V - 1Ah battery through a digital voltmeter. When the battery voltage drops to a low level, the driver presses a key and a focusing flash light fitted at the rear of the vehicle begins blinking at a fixed frequency. This blinking light is the distress signal. It carries no complex data; it simply tells the vehicle behind that help is needed. At the rear part of the front vehicle, a power receiving coil is connected to an LC circuit, a bridge rectifier made of four diodes, and a filter capacitor are used. When a transmitting coil is brought close to this receiving coil, then the electromagnetic field induces a voltage across the LC circuit, the rectifier will converts the AC voltage to DC, and the resulting current flows into the battery. The front vehicle also carries a DC motor driven through a relay and optocoupler, giving it the ability to move along the track under microcontroller control.

The rear vehicle is the answering half of the system. It carries a 12V - 7.5Ah sealed lead-acid battery as its energy source, giving it substantially more stored energy than the front vehicle. At its front face, a small solar panel picks up the blinking light from the front vehicle. An LM324 op-amp comparator converts the panel's varying output into a clean digital pulse train and feeds it into the rear vehicle's own 89C52 microcontroller. The 89C52 decodes the blink pattern, displays the corresponding message on a 16x2 LCD, and switches on a relay that connects the 12V battery supply to the power transmitting circuit. This transmitting circuit is a push-pull oscillator built around two Z44 N-Channel power MOSFETs wired in a center-tapped configuration. The MOSFETs switch alternately at around 30 kHz, driving current back and forth through a centre-tapped coil mounted at the front face of the rear vehicle's chassis. The alternating current in this coil creates an oscillating electromagnetic field that couples into the receiving coil on the front vehicle and pushes charge into its battery.

While the charging is active, a separate safety circuit on the rear vehicle monitors the gap between the two vehicles at all times. An IC567 tone decoder generates a modulated IR beam through a BC557-driven IR transmitter LED pointed forward. If the rear vehicle closes in too far on the front vehicle, the beam reflects off the front vehicle's rear surface and returns to an IR receiver next to the transmitter. The IC567 detects the frequency match and pulls its output pin low. The 89C52 microcontroller reads this as a close detection signal and cuts the motor relay by which the rear vehicle stops until the front vehicle has moved far enough to break the reflection. This stop and start cycle repeats to hold two electric vehicles at the right operating distance through the charging process. The result of this whole work is a self-contained two vehicle system in which the front vehicle signals its need and the rear vehicle detects this signal and responds,[2] energy transfer happens wirelessly between the two electric vehicles and the system is managed safely without any driver intervention.

CHAPTER 2

LITERATURE SURVEY

Wireless power transfer (WPT) has developed a lot over time. It started with the early experiments of Nikola Tesla in 1899, where he showed that energy could be transferred without wires using high-voltage coils. This idea later became the base for modern inductive power transfer systems. In electric vehicles (EVs), WPT helps solve some common problems of wired charging, such as cable damage, difficulty in bad weather, and long charging times that can range from several minutes to hours. With wireless charging, power or the charge can be transferred without physical connections, making the process simpler and more convenient.

One of the main techniques used is resonant inductive coupling. Under this system, the transmitter and receiver circuits are adjusted to the same frequency that is typically 30-85 kHz. Power transfer can be highly efficient, over moderate distances of around 80 90, when tuned correctly. A very similar idea is applied in this project, with a push-pull MOSFET oscillator (with components such as Z44 MOSFETs) converting DC power to high-frequency AC which provides a magnetic field. This sphere causes the coil of reception to increase the voltage and the result is rectified to fill the battery. Vehicle-to-vehicle (V2V) charging builds upon this concept, enabling one vehicle to charge another. In this configuration, the back car is the transmitter and front car is the receiver. This may come in handy when there are no charging stations. According to research studies, V2V charging can be used to alleviate range anxiety among EV users. There are different varieties of WPT systems. Short-range applications may be achieved with near-field inductive systems which have high efficiency and long-range applications may be done with far-field systems with lower efficiency.

In EV applications, resonant inductive coupling is desired as it can accommodate small misalignment and moderate coil-coil distances. Li-Fi is used to complement WPT in the project where rear LED flashing (on-off at low frequency) on the low-battery lead vehicle is used to signal front solar panels on the follower, decoded by op-amps and 89C52 MCUs to LCD messages such as help-battery weak. Light Fidelity (Li-Fi) by Harald Haas in 2011 has 42.8 Gbps through. 200 THz visible/IR spectrum, which is better than Wi-Fi in vehicular

LOS to signal activation. Smart EV reports, per 2025, utilize Li-Fi as a 2-way data, inductive power, to improve coordination during dynamic situations. Collisions during 5-6 cm transfers are avoided by safety features such as close-running detection (IR sensors, IC567). Adoption of EVs is booming (zero emissions, 70 percent cheaper fuel) but requires V2V because of battery drain (e.g., during floods); Phase 2 prototypes have acrylic chassis, DC motors, and two batteries (12V 7.5Ah transmitter, 8 V 1Ah receiver).

The SAE J2954 standards guarantee the interoperability and provide specifications of power (up to 22 kW), frequency (81-90 kHz) and alignment tolerance. The 2024 dynamic WPT at ORNL enhances functionality through segmented pads, which are applicable in V2V motion. Microcontrollers, such as AT89C52 (8051 family, 8K flash) are used in controlling vehicles, maintaining distance, and decoding Li-Fi, and embedded efficiency is provided with Harvard architecture. Efficiency at levels below 70% below 20 cm, high initial cost, and exposure to EMF are some of the challenges, and some of the solutions include ferrite cores, shielding, and compliance with ISO 26262. V2V viability has been demonstrated on a fleet basis (logistics, public transit) with an ROI through fewer downtimes. WiTricity AVs (university prototypes, Toyota dynamic roads) predict bidirectional scalability. Phase 2 gaps will be bridged: mobile bidirectionality across the statical WPT, and Li-Fi to serve real-time requests. Future trends are: kW-scale V2V with AI alignment, V2G, and multi-vehicle networks; Maxa Press 2024 surveys V2V standards to sustainability. IJRASET 2024 notes 90%+ efficiency at kW levels, inspiring Phase 2 enhancements. This survey positions the project as innovative for infrastructure-free EV resilience.

CHAPTER 3

OBJECTIVES AND DESIGN BOUNDARIES

3.1 Scope and Objectives

This project aims to build a two-vehicle system where the vehicles run on electric charge. One vehicle running on high battery charges the other vehicle running on low battery wirelessly with no cables and no stops. The two electric vehicles communicate with each other using Li-Fi technology after which the charging is initiated[3].

1. Design and Construction of Low-Battery Power Receiving Front Vehicle

The foremost task is to build a front vehicle which is meant to receive the charge wirelessly. The construction of front vehicle involves winding the receiving coil, putting the LC circuit and bridge rectifier together, fitting the 8V-1Ah battery with a voltmeter or multimeter and setting up the flash light (Li-Fi Module) at the rear of the vehicle. The 89C52 Microcontroller is used to control the entire system. It monitors the battery level, operates the motor through a relay, and triggers the flashlight to blink when the battery becomes low.

2. Design and Verification of the Push-Pull Oscillator Circuit

The transmitting circuit generates the high frequency electromagnetic field. This circuit uses two Z44 MOSFETs connected in a push pull configuration. It includes 10kΩ gate resistors, BA157 fast-recovery diodes in the feedback path, 470Ω resistors to limit gate current, and 1N4007diodes for protection on drain side. Together these components form a self-oscillating inverter that operates at 30kHz without a need of external clock.

3. Fabrication and Tuning of the Power Transmitting Coil

The electromagnetic field needed to transfer the wireless power between the two vehicles is produced by the transmitting coil[4]. The aim is to engage six turns of 21 SWG wire on a 6-inch diameter ring in a 3+3 centre-tapped system, as shown by the receiving coil. It also seeks to join an appropriate capacitor in parallel with the coil to create an LC circuit and tune the circuit so that it resonates with the frequency of the oscillator. The coil will be positioned on the front of the rear vehicle chassis with the coil facing straight on to the receiving coil of the other vehicle to allow efficient transfer of power.

4. Implementation of the Li-Fi Receiver System

For the rear vehicle to decide when to switch on its transmitter, it needs to detect the blinking light signal sent by the front vehicle. The task is to design the receiver side of this optical link. A solar panel is placed at the front of the rear vehicle to sense the light signal. Its output is given to an LM324 op-amp used as a comparator, with a 680Ω and 1KΩ resistor divider for biasing. This setup converts the varying signal from the panel into a clear digital pulse that the 89C52 microcontroller can read. The microcontroller is programmed to decode the blinking pattern and display the message on a 16x2 LCD. The system is expected to work reliably over the required distance between the vehicles and also handle small changes in alignment during movement.

5. Integration of the IR-Based Proximity Detection System

Maintaining the optimal distance between the two electric vehicles is important for proper operation. If the rear vehicles come too close, there is a risk of collision with the front vehicle, and if it moves too far away then charging becomes ineffective. So, a distance sensing system is designed to solve this problem. IC567 tone decoder with 2.2kΩ and 0.047µF components is used to set the required frequency. A BC557 PNP transistor is used to drive the IR transmitter LED with enough current, and an IR receiver is placed nearby to detect the reflected signal from front vehicle. When the IC567 detects the reflected frequency and pulls pin 8 low, the 89C52 microcontroller should immediately turn off the motor through the relay. When the reflection is no longer detected, the motor should start again. This stop-and-start operation is expected to work continuously and reliably during the movement of the vehicles.

6. System Integration and Real-Time Two-Vehicle Testing

The final objective is integrating the low battery vehicle and high battery vehicle together to complete the entire system for testing and evaluation. The aim is to verify the Li-Fi communication between two vehicles before actual charging process starts, the proper working of the distance detection system and the wireless charging process. This ensures that the signal is transmitted and received correctly, distance between the two EVs is maintained for safety, and the front vehicle with low battery is charged effectively with the help of rear vehicle that transfers charge wirelessly.

3.2 Delimitations

This project is developed to demonstrate that wireless energy transfer between two moving vehicles is possible at a prototype level. It is not intended to be implemented in real vehicles or tested on public roads. The work is limited to a scaled model, and the results are meant to show that the concept works in principle rather than to provide a complete real-world solution.

In this system, one vehicle transfers energy while the other receives it. The project focuses only on two vehicles and does not consider situations involving multiple vehicles or energy sharing in a group. All of the parts of the system are developed separately, with the assumption of the other vehicle.

All the tests are performed on a flat and straight surface and indoors. The system has not been tested on slopes, curvy routes, rugged grounds or in an outdoor environment where several factors such as weather and lighting may interfere with operations. Maintaining the system as simple as it also assists in the clear assessment of how all the components of the system operate.

The batteries used are a 12V, 7.5Ah battery in the rear vehicle and an 8V, 1Ah battery in the front vehicle, which are much smaller than those used in real electric vehicles. Thus, the observed charging rate and power levels in this project are not intended to reflect the performance of EVs in the real world. The speed of the motors is also low enough to be used on a small model.

The two vehicles communicate with each other with a flashlight on the front vehicle and a solar panel on the back vehicle. This creates a one-way communication channel, as the rear vehicle gets the signal but does not respond to it. There are no wireless devices such as Bluetooth or radio communication devices, and a simple one-way signal of the low battery is enough to operate this system.

An IR based detection system is used to control the distance between the vehicles. When the reflected IR signal is sensed (the vehicles are too close) the motor of the rear vehicle is switched off and when the signal is no longer sensed the motor is switched back on. The steering control, speed synchronization and accurate distance measurements are absent; the system maintains a safe distance through this rudimentary stop and start system.

3.3 Constraints

The main physical limitation in this project is the short distance over which wireless charging works properly. The coils can be useful only when separated at a distance between 20 mm and 50 mm. Increasing the distance further than this point, the magnetic coupling becomes extremely weak, and the voltage at the receiving side is too low to charge the battery. It occurs due to the size of the coil and the inherent behavior of magnetic fields in air, and it cannot be changed easily by setting shifts.

The oscillator circuit on the rear vehicle has its frequency determined by the values of the components, approximately 30 kHz. This frequency cannot be changed or regulated as it works. When the frequency varies slightly as a result of heating or variations in the components, the transmitting and receiving coils might not remain well tuned to each other. The consequence of this is that the effectiveness of the transfer of power is diminished and the system does not have a means to identify or rectify this discrepancy.

Li-Fi communication connection can only be effective when the line of sight between the flashlight and the solar panel is clear. When there is something in the line of this signal, it is lost altogether, and the back car is unable to tell whether or not to charge. There are also problems with the system in the bright light conditions where the solar panel itself produces the voltage due to ambient light. This may sometimes reach the comparator threshold when the flashlight is off and the system may not differentiate between real signals and background light.

The distance sensing IC567 tone decoder gives a bare minimum output of a signal that the front vehicle is either near or not. It does not give any information about the exact distance. Due to this fact, the back car will not be able to decelerate progressively but instead, it will suddenly decelerate once the limit is reached. This would need a different kind of sensor to get continuous distance information, and more sophisticated motor control, which is out of the scope of this project.

The 89C52 microcontroller used in the system has limited capability. It has 8-bit processing and runs at 12 MHz, and lacks inbuilt support of more complicated mathematical operations. Simultaneously, it must be capable of performing several functions including the decoding of the Li-Fi signal, tracking of the IR sensor, updating of the LCD, and managing the motor relay. This restricts the use of more sophisticated control procedures that need greater processing capacity.

The coils on the two vehicles are permanent and are unable to move with the vehicles. When the vehicles move laterally, the coils can become out of position, and the coupling between the coils can be decreased. This misalignment is not detected and corrected by any mechanism. The correct alignment is solely dependent on the track and any misalignment can cause a decrease in charging efficiency and have to be manually adjusted to achieve the correct positioning.

CHAPTER 4

PROPOSED METHODOLOGY

4.1 Proposed System

The proposed system is designed to solve the problem of electric vehicle charging during emergency situations by enabling one vehicle to transfer energy to another without using wires. This is achieved using the principle of electromagnetic induction along with a light- based communication method. In this system, two vehicles are involved. One vehicle has a strong battery and acts as the energy provider, while the other vehicle has a weak battery and needs charging. When the battery level of the front vehicle becomes low, the driver sends a request to the following vehicle using a Li-Fi communication system.

Li-Fi works by using light signals to transmit information. [6] A light source in the front vehicle flashes in a specific pattern. This light is received by a solar panel placed on the second vehicle. The solar panel converts the light signals into electrical signals, which are then processed by a microcontroller to understand the message. After receiving the request, the second vehicle activates its wireless power transmission system. The electrical energy from its battery is converted into high-frequency alternating current using an oscillator circuit. The current flows through the transmitter coil, which generates a magnetic field.

When the second vehicle's receiving coil becomes close enough to the above-mentioned magnetic field, electricity starts developing in it. This power is then converted from AC to DC using a rectifying circuit. The resulting power is then regulated and employed for charging the low battery. For security reasons, a distance sensor is utilized. The purpose of employing such sensors is to ensure that there is no collision between two vehicles. With these measures, wireless power transfer and communication is easily achieved.

4.2 Working of Proposed System

The proposed EV-to-EV wireless charging system works by using both wireless power transfer and Li-Fi communication. The whole process happens step by step with both of the vehicles working together. In the beginning, both vehicles run using their own rechargeable batteries. The control of each vehicle is handled by a built-in controller and the 89C52 microcontroller which takes care of communication, movement, and power transfer. When the battery level of the front vehicle becomes low, the driver initiates a request using a control key. This action activates a light source mounted on the vehicle. The light source transmits data in the form of rapid ON and OFF signals, which represent a digital message. This forms the Li-Fi communication link between the two vehicles. The rear vehicle is equipped with a solar panel that acts as a receiver for the light signals. When the transmitted light falls on the solar panel, it generates a small electrical signal corresponding to the light intensity variations. This signal is weak and may contain noise, so it is passed through an operational amplifier circuit. The amplifier improves the signal strength and converts it into a proper digital form.

The processed signal is then sent to the microcontroller in the rear vehicle. From the program, it understands that a charging request has come. After confirming this, the system turns ON the wireless power transmission unit. The transmission part uses a self-oscillator circuit with power MOSFETs. This circuit changes the DC from the battery into high-frequency AC. This high-frequency signal helps in creating a magnetic field for wireless power transfer. The AC output is then given to the transmitting coil. When current passes through this coil, it produces a changing magnetic field around it. When the receiving coil of the front vehicle comes close to this field, voltage is generated in it due to electromagnetic induction.

The voltage generated in the receiving coil is AC in nature. This AC output is then passed through a rectifier circuit, which converts it into DC. A filter capacitor is used to smooth the output and give a more steady and stable DC supply. The DC voltage obtained is then given to the battery of the front vehicle for charging. The charging continues as long as both coils are placed properly and are within a short distance. To check the charging process, a digital voltmeter is connected across the battery. It is also advisable to observe the rise in battery voltage during the process of charging. At the same time, the movement of both vehicles is controlled using DC motors connected through relay circuits. The rear vehicle is set to follow the front vehicle at a controlled speed.

For safety, we used infrared sensors to check how far the two vehicles are from each other. These sensors keep tracking the distance continuously. If the vehicles come very close, the sensor sends a signal to the control unit. Then the microcontroller stops the rear vehicle for a short time using a relay. After the distance becomes normal again, the vehicle starts moving.

In both vehicles, we used voltage regulators so that all the electronic parts get a proper and steady supply. Hence with this the system works properly without any issues.

Overall, the system mainly does three things at the same time:

- Communication between vehicles using Li-Fi
- Wireless transfer of electrical energy

Because of this working, a vehicle with a low battery can get power from another vehicle without using any wires. This is very useful, especially in emergency situations.

4.3 Block Diagram Explanation

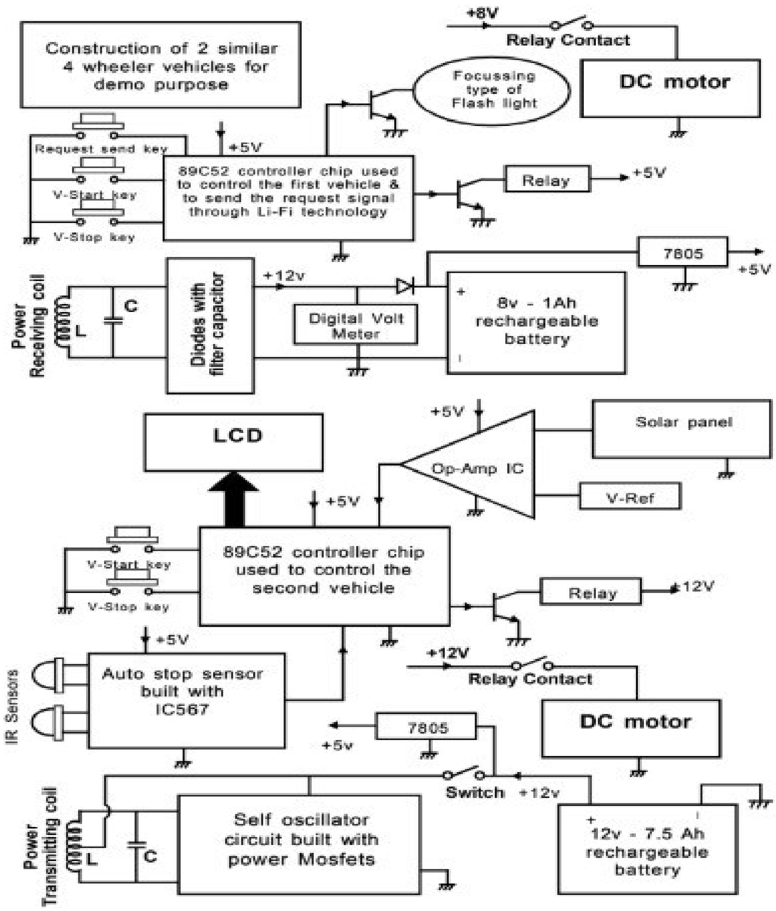

Fig. 4.1 Block Diagram

The block diagram shows the overall working of the EV-to-EV wireless charging system with Li-Fi communication. The system is mainly divided into two parts: the transmitting vehicle (rear vehicle) and the receiving vehicle (front vehicle). Each part includes sections for control, communication, and power transfer.

4.3.1 Front Vehicle (Receiving Unit)

The front vehicle works as the receiving side of the system and it has an 89C52 microcontroller to control the operations. When the battery level becomes low, a signal is sent using a push button. This signal is sent through a light source (flash light) using Li-Fi communication. The front vehicle also has a receiving coil, which takes the energy coming from the rear vehicle. The received signal is in AC form, so it is converted into DC using a rectifier. A filter capacitor is used to make the output smooth. This DC power is then used to charge the battery in the vehicle. A digital voltmeter is connected to check the charging voltage. The movement of the vehicle is done using a DC motor, which is controlled through a relay.

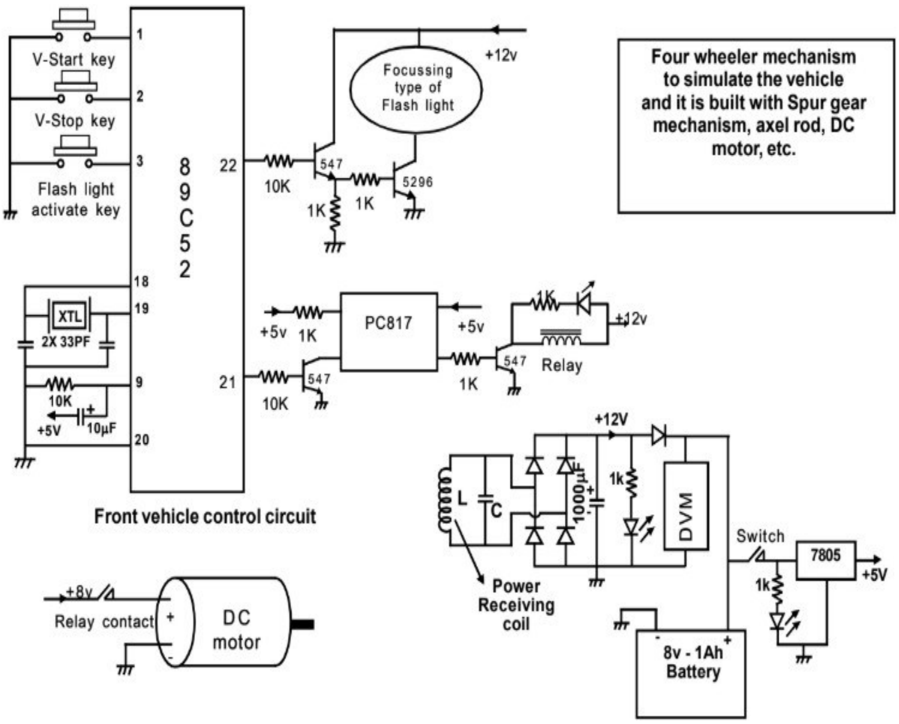

Fig. 4.2 Power Receiving EV Circuit Diagram

4.3.2 Rear Vehicle (Transmitting Unit)

The rear vehicle works as the power transmitting unit and it also has an 89C52 microcontroller. It receives the signal through a solar panel, which converts the light signal into an electrical signal. This signal is then processed using an operational amplifier to make it suitable for the microcontroller. After understanding the signal, the microcontroller turns ON the wireless power transmission circuit. The system uses a self-oscillator circuit with power MOSFETs to change the DC from the battery into high-frequency AC. This AC is then sent to the transmitting coil, which produces an alternating magnetic field.

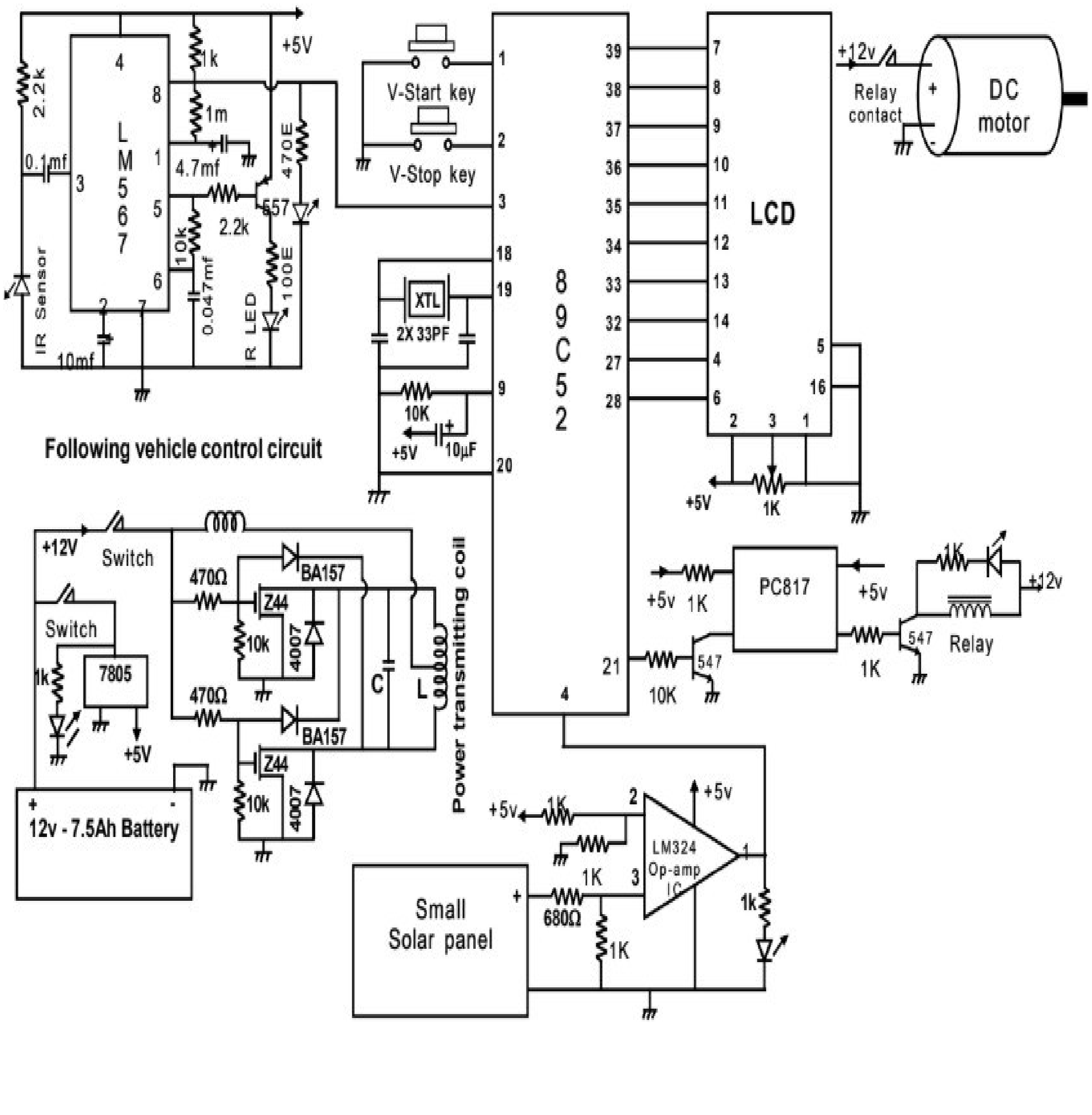

Fig. 4.3 Power Transmitting EV Circuit Diagram

4.3.3 Distance Sensing and Safety Mechanism

To keep it safe, we use IR sensors to see how far the two vehicles are. The sensor signal goes to an auto-stop circuit using IC567. If the vehicles come too close, the circuit sends a signal to the microcontroller, and it stops the motor of the transmitting vehicle for some time. This assists in avoiding collisions and maintaining a proper distance between them.

4.3.4 Power Supply Section

Both the electric vehicles are powered by rechargeable batteries, and while the vehicle at the back uses a larger battery, the vehicle in front uses a smaller one. A voltage regulator such as 7805 is employed for providing a stable +5V power source to the microcontroller and all other components.

4.3.5 Wireless Power Transfer Section

When a high-frequency current passes through the transmitting coil, it produces a magnetic field. [5] If the receiving coil comes into close proximity with this magnetic field, an electrical potential is induced in it as a result of electromagnetic induction.

CHAPTER 5
HARDWARE REQUIREMENTS

The hardware components required for the implementation of the wireless charging of one EV from another using Li-Fi technology are different. They are used for controlling, communicating, transferring power wirelessly, and moving the vehicle. The hardware components can be broadly classified into two categories: transmitter and receiver modules.

5.1 89C52 Microcontroller

Fig. 5.1 89C52 Microcontroller

The 89C52 microcontroller functions as the system's primary controller. It is responsible for controlling and managing all processes and coordinating the activities of all components in both cars. The microcontroller constantly monitors the switch inputs, sensor inputs, and Li-Fi data input. Based on the program, it sends control signals to operate relays, motors, and the wireless charging circuit. It also makes sure communication and power transfer happen properly at the same time. The microcontroller operates using embedded C code stored in its internal memory. Due to its fast processing capability and reliability, it is suitable for real-time embedded applications. In this project, it plays a key role in decision-making and system automation. Without the microcontroller, the system cannot function in an organized and controlled manner.

5.2 Power Transmitting and Receiving Coil

The power transmitting coil is one of the most important components in the wireless power transfer system. It is used to generate a magnetic field when alternating current flows through it. This coil is connected to the oscillator circuit, which produces high- frequency AC signals. The magnetic field produced by the coil carries energy through the air without any physical connection. The strength of the magnetic field depends on the current and coil design. In this project, the transmitting coil is placed in the rear vehicle. When activated, it sends energy toward the receiving coil in the front vehicle. The efficiency of power transfer depends on the alignment and distance between the coils. This method helps to transfer power without using any wires hence the system becomes more flexible.

The receiving coil takes the magnetic field from the transmitting coil. When it comes near this field, voltage is generated in it, and this voltage is in AC form. The receiving coil is designed similar to the transmitting coil so that better power transfer can happen. In this project, it is placed in the front vehicle where charging is needed. The received energy is then converted using a rectifier and filter circuit. If the coils are placed properly, power transfer becomes better. The receiving coil is important because it changes magnetic energy into electrical energy. So, charging can happen without using any wires.

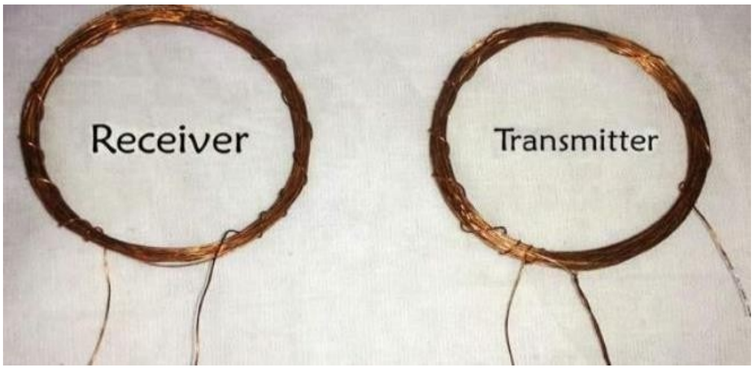

Fig. 5.2 Transmitting and Receiving Coil

5.3 Solar Panel

The solar panel is used as the receiving unit in the Li-Fi communication system. It detects the light signals transmitted by the LED array and converts them into electrical signals. When light falls on the solar panel, it generates a small voltage that varies according to the intensity of light. The change helps to recognize the data that is being transmitted. As the output produced by the solar panel is quite low, it needs to be amplified through an amplifier circuit. The solar panel is sensitive to light changes, which makes it suitable for detecting fast switching signals. In this project, it receives the request signal sent by the transmitting vehicle. The converted electrical signal is then given to the microcontroller for further processing. This enables communication between the two vehicles without using wired or RF methods.

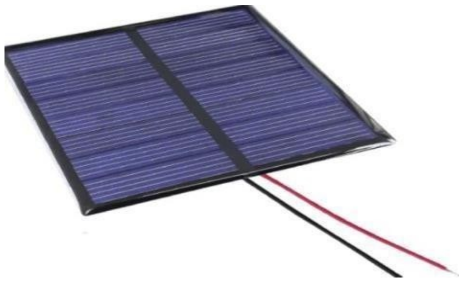

Fig. 5.3 Solar Panel

5.4 Flash Light (Li-Fi)

Fig. 5.4 Li-Fi Module

The LED array is used as the transmitting unit. It consists of the LED which are used to transmit data in the form of light signals by rapidly switching between ON and OFF states. The switching pattern represents digital information that is sent from one vehicle to another.

The use of multiple LEDs increases the intensity of light, making the signal more reliable even under slight disturbances. The LED array consumes less power and responds quickly, which makes it suitable for high- speed communication. In this project, it is used to send a request signal from the front vehicle when the battery level becomes low. The transmitted light is then detected by the solar panel in the receiving section. This method eliminated the need for radio frequency communication.

5.5 Operational Amplifier

The operational amplifier is used to amplify weak electrical signals. In this system, the signal obtained from the solar panel is very small and cannot be directly used by the microcontroller. Therefore, the Op-Amp is used to increase the signal strength. It also helps in reducing noise and improving signal quality. The amplifier ensures that the signal is clear and stable for proper processing. In this project, it acts as an intermediate into a usable form. The Op-Amp improves the accuracy of signal detection in the Li-Fi system. This helps in reliable communication between the vehicles.

Fig. 5.5 Operational Amplifier (Op-Amp IC)

5.6 Power MOSFETs

The MOSFET is used for switching purposes in the circuit. The MOSFET has the ability to switch quickly and also has the capacity to bear high currents. In this project, the MOSFET converts the direct current produced by the battery into alternating current. The alternating current is required to produce a magnetic field inside the transmitting coil. MOSFETS keep switching on and off continually to ensure the working of the circuit. It does not consume any significant amount of power in the process. The functioning of the whole wireless transmission depends on the proper functioning of the MOSFETs.

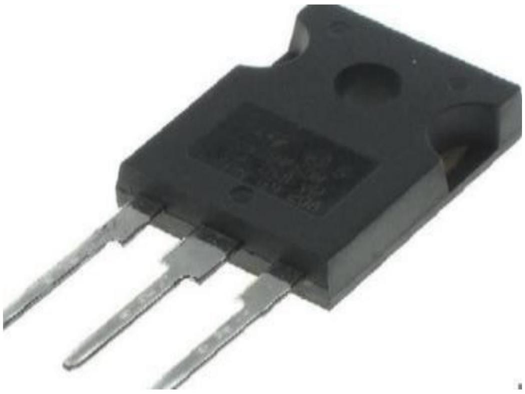

Fig. 5.6 Power MOSFETs

5.7 DC Motor

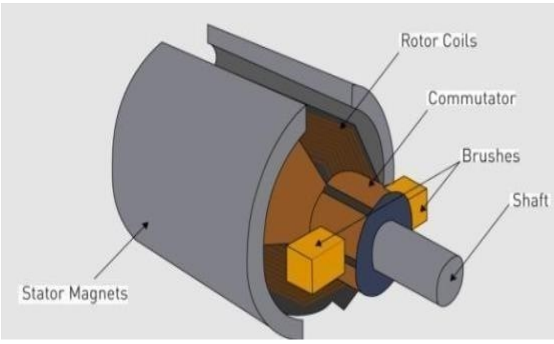

Fig. 5.7 DC Motor

DC motor is used in the model for providing movement to the vehicles. It is a device that changes electric energy to mechanical energy. The motor is controlled using relays and signals from the microcontroller. In this project, the motor is used for simulation of movement of the vehicles on the road. Speed of the motor can be regulated according to the input requirements. Also, it operates in conjunction with the safety system in case the car has to be stopped. Motor is very crucial in demonstrating the practicality of the operation of the system.

5.8 Relay

Fig. 5.8 Relay

Relay is utilized as a switch in the circuit. Relay is required to amplify small signals from the microcontroller to operate large components. Relays are implemented in this experiment to operate the motor circuits and switches. The signal from the microcontroller triggers the relay to turn ON and connect the circuit. Relay provides insulation between the control section and power section to safeguard the circuit elements from high voltage. Relays are easy to use and are found in various circuits. Relays are essential to ensure proper control of operations.

5.9 IR Sensor

The distance between the two cars is measured using IR sensors. These sensors emit an infrared signal, which gets reflected back, and according to that, the distance is estimated. In this project, IR sensors have been mainly employed to ensure safety purposes. The cars maintain a safe distance while moving. If at any point the distance decreases between the two cars, then the IR sensor will send a message to the microcontroller, and the system will stop the motor to prevent the collision.

Fig. 5.9 IR Sensor

5.10 IC567 (Auto Stop Circuit)

IC567 is used as a tone decoder and signal detection component in the system. It is capable of identifying specific frequency signals and producing an output when the required condition is met. In this project, it is mainly used as part of the auto-stop mechanism to enhance safety. The IC works along with IR sensors to detect the distance between the two vehicles. When the vehicles come too close, a signal is generated and processed through the IC567 circuit. Based on this detection, the IC sends a control signal to the microcontroller. The microcontroller then stops the motor of the rear vehicle using a relay. This prevents collision and ensures safe operation of the system. The IC567 provides reliable detection and quick response, which is important for real- time applications. It improves the overall safety and automation of the project.

Fig. 5.10 IC567

5.11 Voltage Regulator

Fig. 5.11 Voltage Regulator (7805)

The 7805-voltage regulator is used to get a constant +5V supply. Components like microcontroller, sensors and ICs need proper fixed voltage to work. If voltage changes, they may not work correctly and sometimes can get damaged. In our project, battery voltage is more and not stable. So we use 7805 to bring it down and keep it steady. Because of this, all components work properly. It also helps to reduce noise and sudden voltage changes. Without this, the system may not work correctly. So, 7805 is important for smooth working of the system. Thus, the 7805 is essential for safe and efficient operation.

5.12 Rechargeable Batteries

Fig. 5.12 Battery

Rechargeable batteries are used as the main power source for both vehicles in the system. They store electrical energy and deliver electricity to all components of the device during operation. For this project, two different kinds of batteries are employed depending on the necessity. Since there is a need to provide power for transmission from the transmitting vehicle, therefore, a larger size battery is needed in order to accomplish that objective. The other kind of battery that is used in the receiving device charges during the operation. The batteries can be recharged many times, hence increasing the efficiency of usage and reducing costs. The performance of the battery affects how the system works. Choosing the correct battery helps the system run for a longer time. So, batteries are very important for running the whole system.

5.13 Rectifier (Diodes)

Rectifier circuit is used to change AC into DC. The power coming from the receiving coil is in AC form, but battery needs DC to charge. So, we will be using a rectifier with diodes. These diodes allow the current to pass only in one direction. Hence, AC gets converted into DC. In this project, this helps us use the received power for charging the battery. Without this, the power we get cannot be used properly. So, rectifier is an important part in the charging section. The rectifier improves the ability of the transmitted energy. It is an essential part of the charging circuit.

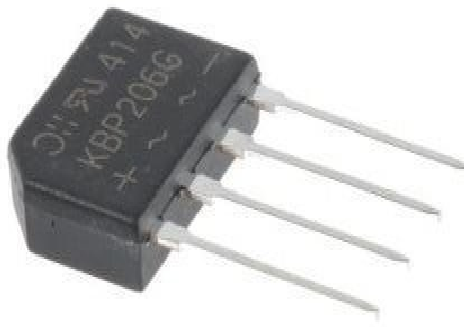

Fig. 5.13 Rectifier

5.14 LCD Display

The LCD display is used to show the information to the user. It shows the main details about how the system is working. In this project, the LCD is interfaced with the microcontroller, displaying information such as the system's status and signals received. This enables us to determine whether the system is communicating and charging effectively. The LCD is user-friendly and requires very minimal energy. Different messages can be displayed based on the program used. This enhances the usability of the system and makes it more informative. It also plays an integral part in testing and debugging.

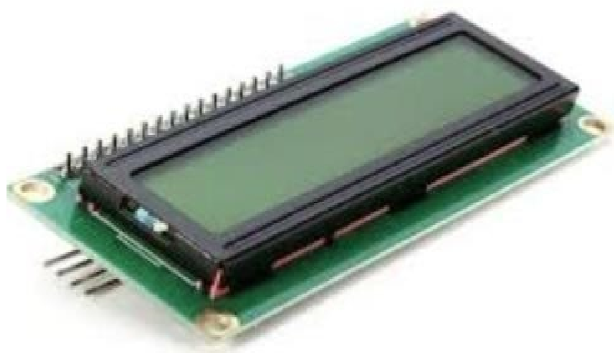

Fig. 5.14 LCD Display

5.15 Digital Multimeter

A digital multimeter is an instrument which measures the voltage, current, and resistance of a circuit. During our experiment, we made use of it to test the charging process of the voltage of both vehicles. We connected the probes at the terminals of the battery and observed the results. We were able to observe the increase in the voltage during the charging process, which showed us that there was power transmission taking place.

Fig. 5.15 Digital Multimeter

5.16 Switches and Control Keys

Switches and control keys are used to operate the system manually. They help the user control different functions easily. In this project, switches are used to start and stop the vehicles. A special key is used to send the Li-Fi signal from the front vehicle. These inputs go to the microcontroller, which acts based on them. Switches make it easy to control the system. They are simple and easy to use. They also help while testing the system. Without switches, we cannot control the system manually. So, they act as a link between the user and the system.

5.17 Connecting Wires and PCB

Connecting wires and PCBs are used to join all the components in the system. They help in proper flow of signals and power. In this project,wires are connected to sensors, microcontrollers, batteries, and other parts. The PCB gives a space to fix all the components. It makes the circuit neat and easy to understand.

It is necessary to have good connections to ensure that signals are not lost. Good connections will make the system reliable. It also makes the process of troubleshooting easy. The PCB also makes the system robust and organized. Therefore, all these components are vital to the whole system.

Fig. 5.16 Connecting Wires

CHAPTER 6
SOFTWARE REQUIREMENTS

6.1 Embedded C Software

The software aspect of the EV-to-EV wireless power transmission system is implemented using Embedded C programming. It is well suited for embedded microcontroller projects because it gives full access to the hardware modules. In this project, Embedded C is employed to write the code necessary for communication, switching functions, and system integration. The software will be saved in the microcontroller memory and run continuously to control its operation.

Embedded C is utilized in this project to:

- Sample the input data from switches and sensor arrays
- Operate the LEDs with a blink sequence for Li-Fi signal transmission
- Process the input data from the solar panel
- Cycle the relays to switch ON the chargers and motors
- Ensure synchronization among the transmitter and receiver systems

6.2 Keil Software

The software Keil is used to write and compile the Embedded C code. All the tools to develop, compile and debug are provided by this software. The code developed through the software Keil is compiled into machine code or hex code which can be used to load into the microcontroller. Keil provides a user-friendly environment where one can write and modify codes conveniently. In addition, there is a built-in compiler to detect any syntax error in the program ensuring that the program is free from any error prior to its execution. It facilitates debugging which enables testing the program in a step-by-step manner to locate the logical errors, if any. There is also an option for simulation of the program to validate it without any hardware connection.

Keil software is used for this project to:

- Write the Embedded C program for 89C52 microcontroller.

- Compile and debug the program.
- Generate hex code for microcontroller programming.

6.3 Microcontroller Programming (Flash Programming)

The next step involves uploading the program into the microcontroller using a programmer tool. This is referred to as flashing. The hex file produced through Keil is uploaded to the microcontroller's memory. After programming, the controller operates by itself.

Some applications of programming in this project include:

- Storing the control algorithm within the microcontroller
- Making the circuit react automatically to the
- Operating the relays and motors according to specified conditions
- Ensuring the system runs continuously

Programming is very important since it makes the hardware perform its intended purpose.

6.4 Signal Processing Logic

This model features some basic signal processing for interpretation of the Li-Fi signal used in communication. The light signal picked up by the solar cell is changed into an electric signal. This signal will then be processed through the microcontroller to detect if there is any communication request made. The light signal picked by the solar cell is changed to electric signal form, but normally, it is of low strength. The signal can have notices due to environmental factors.

The signal will then be fed to the microcontroller, whose logic involves:

- On and off detection of light signal
- Converting analog signals to digital inputs
- Making comparison between detected signal and pre-defined parameters

6.5 Control Logic for System Operation

The system functions depending on the predetermined criteria programmed into the system. Once there is a signal for the request, the microcontroller will switch on the wireless charging system. Simultaneously, the microcontroller regulates the functioning of the motor through sensor feedback.

The control logic includes:

- Checking input conditions continuously.
- Activating relays for power transmission.
- Controlling DC motor movement.

Stopping the vehicle when distance is low (IR sensor input) This logic ensures smooth and safe operation of the system.

6.6 Debugging and Testing

Testing and debugging are important steps to ensure proper functioning of the system. During development, the program is tested multiple times to verify correct signal transmission, reception, and power transfer.

Debugging helps to:

- Identify and correct logical errors.
- Verify communication between modules.
- Ensure proper switching of relays.
- Confirm stable operation of the system.

CHAPTER 7
SYSTEM IMPLEMENTATION

The integration of wireless power transfer and optical communication between two electric vehicles is the main goal of the system implementation of the suggested EV to EV wireless charging with Li-Fi based communication system. The goal is to use light signals to facilitate vehicle-to-vehicle communication while transferring electrical energy without physical connections. Two prototype cars are used to develop the system; one serves as the transmitter and the other as the receiver. Both vehicles have a number of subsystems, including wireless power transfer, Li-Fi communication, and distance sensing, and are managed by AT89C52 microcontrollers.

7.1 Transmitter Unit Implementation

The rear vehicle, which serves as the energy source, has the transmitter unit installed. Powered by a rechargeable battery, it has a transmitting coil, power MOSFETs, and an oscillator circuit. In order to facilitate wireless power transfer, the oscillator circuit transforms DC power into high-frequency AC. When a Li-Fi signal is received from the front car, the microcontroller activates the transmitter and regulates its operation.

The main components of the transmitter unit include:

- Rechargeable battery (power source)
- MOSFET-based oscillator circuit
- Transmitting coil
- Microcontroller (AT89C52)

Circuit Connections (Transmitter Unit):

The Transmitter Unit is basically an electric vehicle used to transmit charge to another electrical vehicle with weak battery. A push-pull arrangement of MOSFETs is used in the design of the transmitter circuit. The circuit receives DC voltage from the battery, and the MOSFETs alternately switch to produce high- frequency AC signals. The MOSFET outputs are connected to a center-tapped coil, and resonance is maintained by an LC circuit.

Working of Transmitter Unit:

The transmitter unit operates by converting electrical energy into electromagnetic energy. The oscillator circuit is turned on by the microcontroller upon receiving a request signal via Li-Fi. The transmitting coil receives the alternating current produced by the MOSFETs' high-frequency switching. As a result, the magnetic field surrounding the coil varies. The following is a summary of the working process:

- The front car receives the Li-Fi signal.
- The transmitter is activated by a microcontroller, and MOSFETs produce high-frequency AC
- A magnetic field is produced by a transmitting coil. Wireless energy transfer when both coils are positioned closely together (about 3–5 cm), efficient power transfer takes place.

7.2 Receiver Unit Implementation

The front car, which has a weak battery, is where the receiver unit is located. It is in charge of gathering wireless energy and transforming it into electrical power that can be used. A receiving coil, rectifier circuit, filter capacitor, battery, and Li-Fi receiver system make up the unit. Among the receiver unit's crucial parts are:

- Bridge rectifier with receiving coil
- Capacitor filter
- battery that can be recharged
- Li-Fi receiver (op-amp + solar panel)

Circuit Connections (Receiver Unit)

The receiving coil is connected in parallel with a bridge rectifier circuit. This rectifier circuit converts the AC voltage generated in the coil into DC voltage. A capacitor is connected in parallel with the rectifier circuit for filtering the DC voltage. This DC voltage is then used for battery charging. The Li-Fi receiver circuit consists of a solar panel connected in parallel with an op-amp (LM324), which is a comparator.

Working of Receiver Unit:

The receiver unit operates on the principle of electromagnetic induction. When the receiving coil is brought within the magnetic field generated by the transmitting coil, an induced voltage is obtained. This voltage is used to charge the battery.
The steps involved in this operation are as follows:

- Receiving coil detects magnetic field
- Induced AC voltage is generated
- Rectifier converts AC to DC
- Capacitor filters DC voltage
- Battery is charged

During this time, the Li-Fi receiver also receives light signals and forwards them to the microcontroller.

7.3 Li-Fi Communication Implementation

The Li-Fi communication system facilitates data transmission between two vehicles using visible light. In this system, an LED is used as a transmitter that transmits information using light. A solar panel is used to receive this information.
The steps for communication are as follows:

- LED is used to transmit information using light
- The information is received using a solar panel
- The information is converted to a digital signal using an op-amp
- The message is decoded using a microcontroller

7.4 Close Distance Detection System

To prevent collision between vehicles, a distance detection system is implemented using IR sensors and IC567. In this system, the IR transmitter is continuously sending signals. The IR receiver receives these signals and detects the reflected signals coming from the object. Once the distance between vehicles is minimized, a signal is sent and processed by the microcontroller to stop the vehicle temporarily.

7.5 System Integration

The complete system is made up of the integration of various modules such as the wireless power transfer module, Li-Fi communication module, microcontroller module, and distance sensing module. All the modules have a unique functionality and integrate with each other in a coordinated manner.

The integrated system consists of the following modules:

- Power transmission module
- Power receiving module
- Communication module
- Safety module

The microcontroller is the central controlling unit of the system.

CHAPTER 8
TESTING AND EVALUATION

The testing and implementation of the proposed EV to EV wireless charging system with Li-Fi communication is carried out in order to test the functioning of the proposed system. In the proposed system, the various modules include the wireless power transmission and reception circuit, Li-Fi communication module, microcontroller module, and close running detection module. All the modules in the proposed system are tested individually and then integrated in order to test the functionality of the proposed system. The aim of the test is to transfer electrical energy wirelessly and communicate between the vehicles.

8.1 Testing Methodology

The testing methodology is carried out in a systematic way by dividing the system into various sections. In the beginning, each circuit such as the power transmitter circuit, receiver circuit, Li-Fi communication circuit, and sensing circuit is tested separately. After the testing of each circuit is complete, the overall system is assembled and tested in real-time conditions.

The overall testing methodology of the system includes the verification of the connection of the circuit, checking the response of the microcontroller, and checking the output of each module of the system. The system is tested with various distances between the coils and under various conditions of light.

8.2 Sensor Testing

The close running detection circuit uses IR sensors and IC567 tone decoder for the test. The IR transmitter sends out infrared signals at all times, and the IR receiver receives the reflected infrared signals from the front vehicle. During the test, it has been seen that in the absence of a front vehicle, the IR signal is constantly sent and received, maintaining a stable state. However, as the front vehicle approaches the current vehicle, the state of the IR signal changes, causing the tone decoder to change state. This signal is sent to the microcontroller, which stops the vehicle momentarily to keep a safe distance. This confirms that the circuit has been designed to prevent accidents and ensure the safe passage of vehicles.

8.3 Li-Fi Communication Testing

The communication of the Li-Fi system is tested by transmitting signals through an LED source and receiving them through a solar panel [7]. These received light signals are then allowed to fall on a solar panel, producing electrical signals. These electrical signals are then passed through an op-amp, a comparator, where they are compared. Depending on this comparison, a microcontroller decodes the signal and displays it on an LCD. During the communication test, it is seen that the message "PLEASE HELP ME MY BATTERY IS WEAK" is displayed on the LCD.

8.4 Power Transfer Testing

The wireless power transmission system is tested by using a high-frequency oscillator circuit to supply power to the power transmitting coil. The electromagnetic field developed by the transmitting coil induces a voltage in the receiving coil. The induced AC voltage is converted into a DC voltage using a bridge rectifier circuit. A capacitor is used as a filter to obtain a smooth output. A digital voltmeter is used to measure the output voltage. During testing, it is observed that:

- Approximately 10V to 12V DC is obtained at the receiver side
- Current of 200-300 mA is obtained
- The weak battery is charged

The efficiency of the power transmission is high when the coils are kept parallel and the distance between the coils is less than 5 cm. As the distance between the coils increases, the efficiency of the power transmission decreases.

8.5 Integrated System Testing

The next step is testing the system as a whole. This is done using an integrated model. This is done when the front vehicle sends a request signal using Li-Fi communication. This signal is sent to the rear vehicle. This signal is used to activate the power transmitting circuit. The transmitting coil is used to produce an electromagnetic field. This electromagnetic field is used to charge the battery using the receiving coil. Meanwhile, the IR sensing circuit is used to keep the distance between the two vehicles.

8.6 Validation

The system can be validated by comparing the expected performance with the results obtained during testing. The results show that the system performs as expected. The results obtained from the validation are:

- The transfer of power wirelessly was successful.
- The Li-Fi communication was accurate.
- The system responds to the control signals appropriately.
- The prototype was feasible.

8.7 Error Handling

During testing, certain limitations and errors are found. These problems include misalignment of the coil, difference in light intensity, and improper connections. In order to address these problems, the following actions can be taken:

- Proper alignment of the coil to improve efficiency.
- Testing in proper lighting conditions.
- Proper connection.

This will help improve efficiency and reliability.

CHAPTER 9

RESULTS AND ANALYSIS

This chapter discusses the results obtained by implementing the EV-to-EV wireless charging system with Li-Fi-based communication. The performance of the different modules of the system, namely the wireless charging, Li-Fi communication, and distance sensing, has been evaluated on the basis of the experimental observations made on the system.

9.1 Experimental Setup

The test configuration is composed of two prototype cars acting as the transmitter and receiver. The two cars are placed side by side to make sure that the coils align properly.

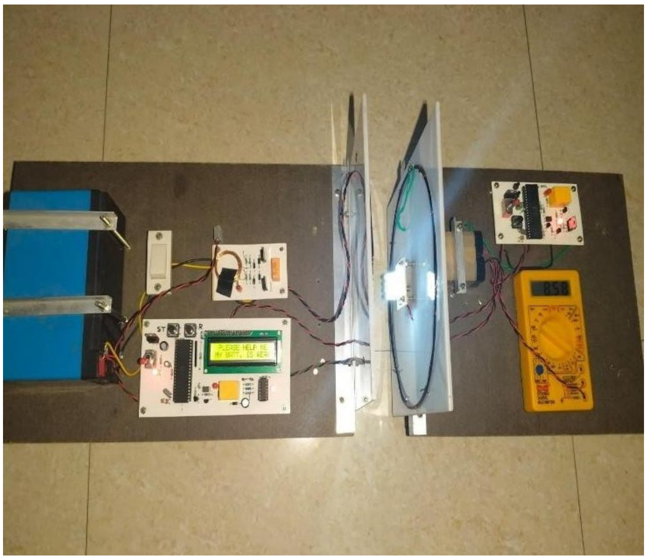

Fig. 9.1 EV-to-EV Charging System Using Li-Fi Communication – Hardware Prototype

9.2 System Results

Testing of charging was done using the Li-Fi communication channel where the signals were transmitted using an LED light and received using a solar cell. From the test, it was evident that Li-Fi technology was capable of detecting variations in light intensity.

Results that have been recorded are:

- Decoding of messages through the microcontroller
- Transmission of signal via light pulses
- Communication via indoor lighting

When the testing stage was reached, the device was powered up and the communications channel was set between the two cars. Pressing the start button on the transmitting car triggered the flashlight LED to start emitting modulated light pulses at the solar cell of the receiving vehicle. These were received and decoded properly by the 89C52 microcontroller in the receiving car, resulting in a visible output on the LCD screen: "Message received".

"PLEASE HELP ME MY BATTERY IS WEAK"

This meant that all of the parts of the Li-Fi signal process, including the transfer of light, decoding of the message, and display of the message on the screen, worked as planned.

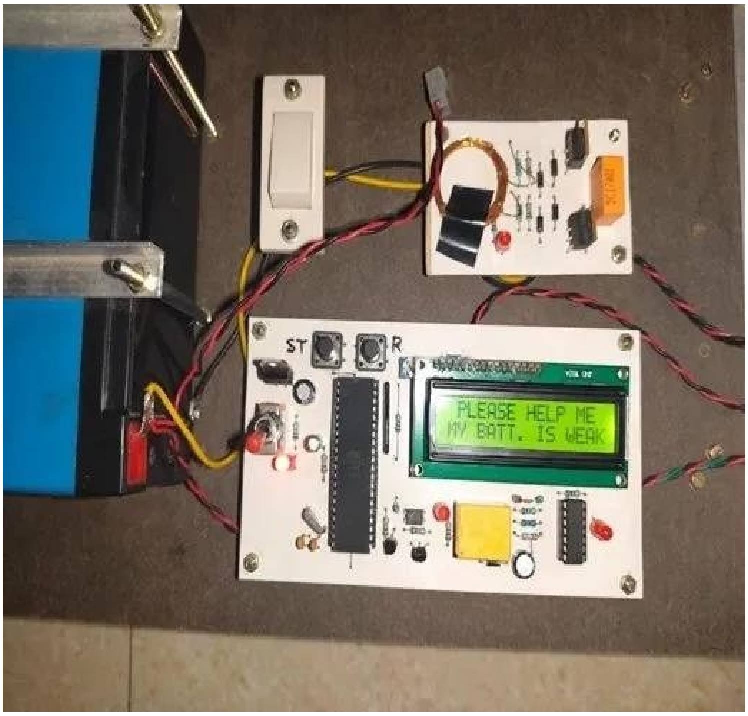

Fig. 9.2 Message of "Low Battery" through Li-Fi Communication

Alongside this, the digital multimeter connected across the receiving circuit registered a voltage of 8.72V, which validated that the wireless power receiving coil, along with the rectifier and filter stage, was successfully converting the transmitted energy into usable DC voltage. This proved the working of the battery charging side of the project too, and that it was providing a constant output.

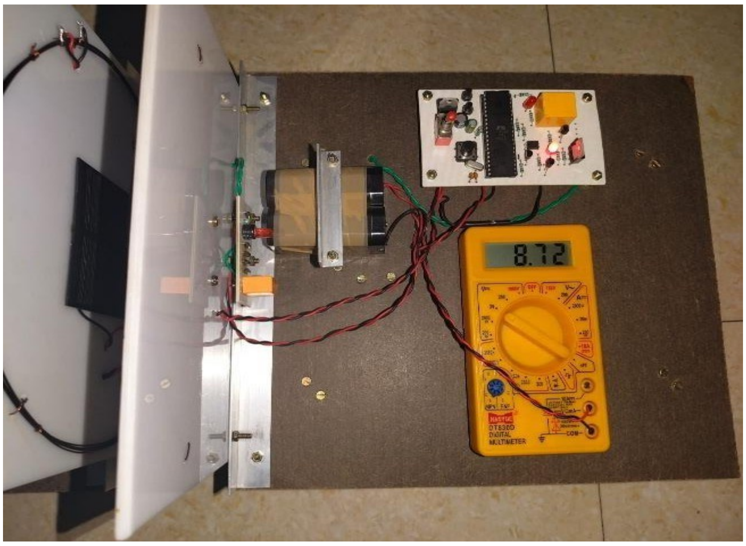

Fig. 9.3 Measuring Battery Charge Using Multimeter

9.3 Performance Analysis

The performance of the system is evaluated based on the performance of power transmission, communication, and response.
The following observations are recorded:

- High efficiency is obtained when the distance is short (3–5 cm)
- Efficiency decreases when the distance is increased
- Communication is stable using Li-Fi
- Microcontroller response is quick in activating the system

CHAPTER 10

APPLICATIONS AND ADVANTAGES

10.1 Applications

The designed system of EV-to-EV wireless charging integrated with Li-Fi communication has several potential applications in the modern transportation and smart mobility context. This technology could be used in various practical scenarios where emergency charging, smart coordination, and cable-less power transfer are necessary.

1. Emergency On-Road Charging Assistance

The proposed system can provide emergency roadside charging support. In situations where an EV depletes its battery on a remote road, highway or in a traffic jam, a nearby EV can offer wireless power. This is helpful in avoiding vehicles break-down and safely drive the battery depleted EV to the closest station.

2. EV-to-EV Power Sharing

The proposed system enables electric vehicles to share energy among themselves. In this scenario, an EV may act as a donor and the other as a receiver, which is helpful where charging infrastructure is not available or in order to help a vehicle in immediate need. It supports a cooperative energy ecosystem among the EVs and makes electric transportation reliable.

3. Smart Fleet Management

This system can be beneficial for electric buses, taxis, commercial vehicles, and delivery vans. Electric vehicles with a high battery can assist those with low battery, which enhances their operational usage and reduces downtimes in fleet management, such as in logistics and public transportation.

4. Dynamic & Mobile Charging Systems

The project indicates that charging can occur at a distance and while moving. This principle could potentially be developed into a dynamic charging system where electric vehicles are constantly transferring energy while in motion, which may help decrease reliance on fixed charging infrastructure.

5. Smart Road and Intelligent Transport system

Integrated into the smart road, smart highway & smart connected mobility infrastructure the smart road would be able to support a plethora of features including prompting for emergency charge; showing charging details based on traffic; enabling within car communications with Li-Fi etc.

6. Safety in Harsh conditions

Being a wireless system the device would operate in many of the aforementioned conditions, including rainy, wet/water logged areas, industries with hazardous/dusty conditions, and even areas where open-wire charging is riskier.

7. Advancement of research and prototypes

The project would be a prototype for advancement in research in several key areas including wireless power transfer, Li-Fi communications, embedded systems, automatic electric vehicle charging and smart transportation.

10.2 Advantages

The proposed smart charging system provides several benefits over traditional wired EV charging methods

1. Wireless charging (no wires used for charging)

It is an advancement over the wired charging system as a physical connection (wire, plug, unplugs etc) is not needed in this case.

2. Aids in emergency

The electrical vehicle that is working fine, can be used to charge the non-working vehicle during a breakdown with less anticipated charging and therefore a user can recharge his vehicle by docking his charged electrical vehicle with a non-working electrical vehicle and it will start charging. On a highway it helps as charging station are far away.

3. Safety

Significant benefits are achieved with the elimination of a physical connection.

- no user gets electrocuted.
- no electrical shock or electrical arcing is occurred.
- The system can be water proof.

The electrical vehicle charging port interface doesn't gets worn out since there is no sparking while electrical vehicle port insertion.

4. High speed data transfer with Li-Fi based charging

- Data transfer at high speed takes place.
- Very small amount of electrical charge consumed during part charging since very less amount of energy is utilized during data transfer.
- Less interference as Li-Fi based data transfer is used compared to radio-based frequency data transfer.

5. Immunity from wireless interferences

- Like Bluetooth or Wi-Fi R/F signals would be used for communication and information transfer.
- There would be no issue of RF interference anymore.
- Information would be wirelessly transmitted to and from the car at a short distance, so that it is done reliably and thus the security risk is also minimal.
- Communication between cars with safety assured at short range can be done.

6. Constant maintenance of distance automatically

It would be ensured using IR based proximity sensors that an almost constant range is maintained between vehicles and collisions do not occur and that charging coils are always properly aligned.

7. Lesser wear and tear of mechanical parts

- The mechanical system used to plug and unplug sockets and connectors for the charging of the electric vehicle would be rendered obsolete.

- Wear and tear of mechanical parts will be less to an optimum level.
- Less maintenance costs would be incurred.
- Increased reliability of parts for future.

8. Future applications for Smart EV system

- Cooperative EV Charging
- Smart Transportation Network.
- Fully compatible for autonomous EV systems.

9. Compact and Simple Embedded Design

- AT89C52 microcontrollers
- solar panel-based Li-Fi receiver
- LM324 comparator

CHAPTER 11
FUTURE SCOPE

The proposed system proved that the wireless data, EV to EV wireless power transfer could be feasible when applying it in a practical case study by some ways, in the proposed system Smart Charging with Li-Fi and EV to EV Wireless Charging. But the implementation prototype is only a prototype and the research could be started from it for applying into the real application case study as below:

1. High-power Wireless Charging Implementation

The wireless power transfer of low power and short distance have been implemented into the prototype and the implemented prototype is capable of implementing into the practical case study. To transfer power for high-capacity battery for EV.

- High power resonant inductive coils.
- The proper compensation network.
- The optimized inverter and converters design.
- The adequate shielding and thermal control design.

2. Higher Transfer Distance

The wireless power transfer distance of few cm. The issues to be design with a higher transfer distance system is to:

- Increased size of resonant coils
- Increased the number of turns of the resonant coils
- Matched resonant frequencies
- Using a magnetic core for coil.
- The mechanism of EV positioning

3. Bi-directional EV to EV charging

Currently one EV will be the transmitter and the other an receiver, but a state where both the EVs can be constructed as a bi-directional EV to EV charging unit and both the EVs can be:

- Transmitting power
- Receiving power
- Assigned as a transmitter/receiver dynamically
- Used as a power source as well when needed shall be achieved.
- This can be part of a smart cooperative and adaptive EV to EV energy sharing network.

4. Advanced Li-Fi communication system

In its current implementation, the system uses a very basic Li-Fi system involving LED based transmitter and solar cells for the receiver. The future enhancements of this system could involve:

- Faster photodiodes
- Improved optical filters
- Enhanced modulation scheme
- Duplex communication capability
- Secure communication protocol
- Real-time control
- immunity to external day light.

5. Interfacing with Battery Management System (BMS)

To be practical in a vehicle this system has to be integrated with a Battery Management System (BMS) to be able to monitor the health and State of Charge (SoC) of the battery, regulate charging based on battery temperature, provide over-voltage/over-current protection and smartly cut off the charging based on certain conditions.

6. Autonomous Vehicle Coordination

In future applications of this system, interfacing of the charging robot with an onboard sensor, an AI-based control unit, an autonomous navigation controller and vehicle-to-vehicle coordination algorithms could be done to let the assisting vehicle locate the low battery vehicle, automatically drive to a correct position, maintains optimum charging distance and starts the charging process automatically without human intervention. This will be critical for applications with fleets of autonomous EVs and advanced transport systems.

7. Smart roads-dynamic charging

This can evolve from the above systems into smart roads. These will charge while driving, via coils placed in them and controlled using smart Li-Fi signaling for transmitting the charge power. These systems will not suffer from range anxiety, and also reduce the need for a bigger EV battery.

8. IoT & cloud-based monitoring

This can help in effective management and monitoring of smart charging systems through an app on a mobile phone, or a cloud-based dashboard. Features such as remote battery diagnostics, vehicle location & tracking, and generation of vehicle usage and maintenance requirements reports along with predictive maintenance can also be included.

9. Safety and Standardization

Essential consideration will include EMI & FOD shielding, bio-safety protection and standardizations for smart charging systems in cars to allow marketability and interchangeability.

10. Renewable Energy integration

EV to EV charging systems can also be made to be environment friendly by incorporating alternative sources of energy such as portable mobile power sources charged from solar energy, solar charging power bank attached on EVs.

CHAPTER 12
CONCLUSION

In this 'Smart Charging using Li-Fi, EV to EV wireless Charging' project, the working system of wireless charging between EV and also the optical real time communication between EV via Li-Fi technique is demonstrated. As studied in the above sections, the EV (low battery) needs help from its neighbor EV (helping EV) via signal from light communication system and received power from other EV via wireless (resonant inductive charging) from the helping EV. In this project, the system has following sub-systems: A control system with AT89C52 microcontroller, Li-Fi communication system with LED as transmitter and solar cell as receiver, resonant inductive coils, batteries rectification and charging circuit and IR system to identify neighbor EV and maintain proximity.

From the test result as given, system can be trusted in terms of steady range, short range wireless power transfer, Li-Fi communication, EV cooperation. The report mentioned that a steady DC power output of 10-12V and a charge current of 200-300mA was acquired at a short range and this is exactly the problem range anxiety and help when in need for current EV technology.

EV to EV wireless power transfer means the EVs can even help itself without any outer infrastructure to charge. It is the most practical and reliable vehicle when we are using electric cooperative energy sharing system, Li-Fi part has improved system more responsive, secured, cooperative between vehicles and also easy in charging operation. Though the demonstration here is small scale, this study provides potential for the development of large-scale and high-power wireless charging, smart EV network, autonomous vehicles assistance, dynamic charging infrastructure, intelligent battery management systems etc. We can conclude that proposed system is novel, secure.

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